OdinAI: AI-Powered Dental Disease Classification (Project Code:6)

Abstract

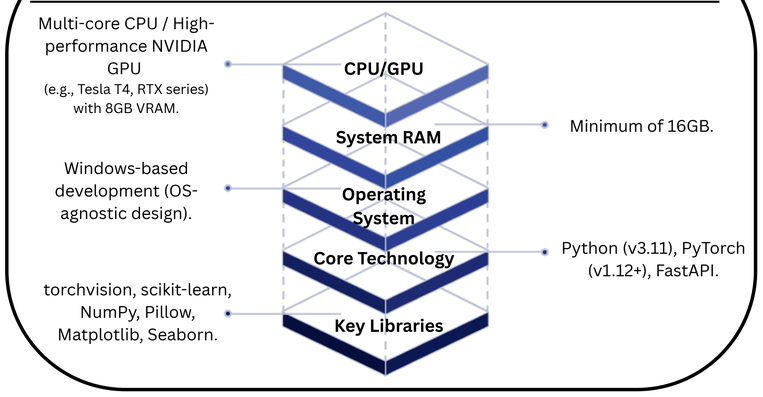
92.28% ***Overall Accuracy***

89.3% ***Macro F1-Score***

1.2s ***Inference Time***

ScanAI presents an advanced AI system for automated dental disease classification using a Vision Transformer (ViT-B/16) architecture. Trained on 6000 professionally annotated dental images across 6 disease categories, the system achieves an exceptional 92.28% overall accuracy with clinical-grade precision.

Experimental Setup



Origin of Research Problem

This research addresses the global accessibility gap in oral healthcare. Challenges include a lack of access for 60% of rural populations, cost avoidance by 70% of individuals, and diagnostic variability of 15–25% among practitioners. The solution is an AI-powered preliminary screening tool to democratize dental health assessment and enable early detection in underserved regions.

Global Crisis: 3.5 billion people affected by dental diseases worldwide

Objectives of the Work

Improve Diagnostic Accuracy: To use a highly precise AI tool to standardize dental assessments and reduce the 15-25% diagnostic variability seen among professional dentists[1].

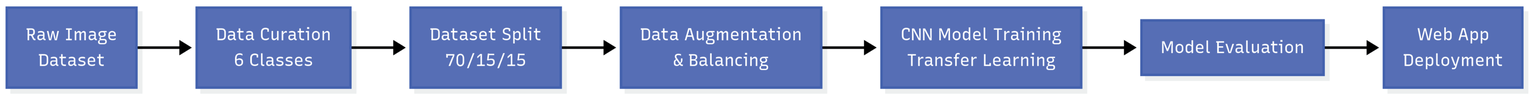
Increase Healthcare Access: To democratize oral healthcare with a fast, accessible web-based screening tool that overcomes barriers of cost and location for early, affordable detection[2].

Build a Complete System: To deliver a production-ready, end-to-end application with a secure backend and user interface, ensuring it is a practical tool ready for real-world deployment[3].

Methodology

A systematic approach was employed, encompassing data preparation, model training, and evaluation. The entire process is visualized below:

Figure 1: Project Workflow

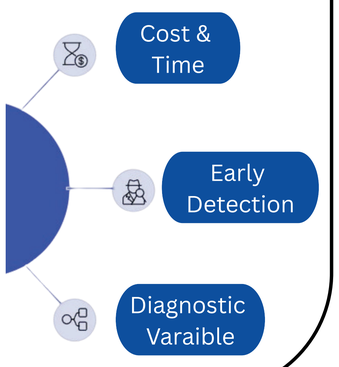


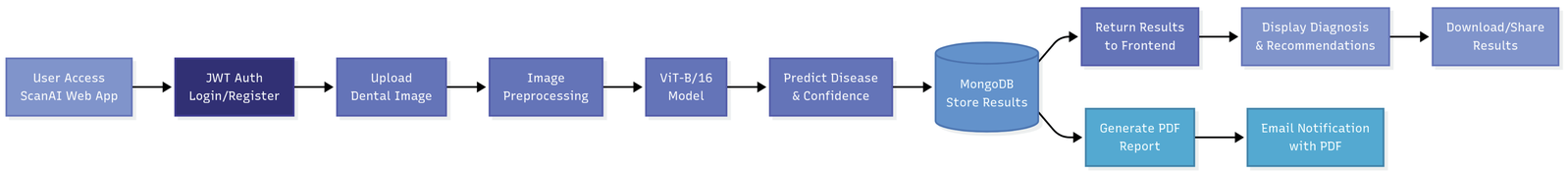
1. Dataset Curation and Preprocessing

- **Dataset**: A collection of photographic images categorized into six classes: Caries, Gingivitis, Calculus, Mouth Ulcers, Tooth Discoloration, and Healthy Teeth.
- **Data Splitting**: The dataset was partitioned to ensure unbiased evaluation:
  a. **Training Set**: 70% (for model learning)
  b. **Validation Set**: 15% (for hyperparameter tuning)
  c. **Testing Set**: 15% (for final performance assessment)
- **Data Augmentation**: The training set was expanded to improve generalization and balance classes using:
  d. **Geometric Transformations**: Rotations, flips, scaling.
  e. **Photometric Transformations**: Brightness, contrast, saturation.

2. Deep Learning Model Architecture and Training

- **Model Selection**: A Convolutional Neural Network (CNN) using transfer learning from a pre-trained architecture (e.g., ResNet50 or EfficientNet).
- **Training Regimen**:
  a. **Loss Function**: Categorical Cross-Entropy.
  b. **Optimizer**: Adam optimizer.
  c. **Monitoring**: Tracked accuracy and loss on training and validation sets to ensure convergence and prevent overfitting.





Project Specification

Component	Specification
Model Architecture	Vision Transformer (ViT-B/16) with a custom classification head.
Input	224x224 pixel RGB images.
Output	6 distinct classes (Calculus, Caries, Gingivitis, Healthy Teeth, Mouth Ulcer, Tooth Discoloration).
Loss Function	Focal Loss (with gamma γ=2.0) to handle class imbalance.
Optimizer	AdamW (Adam with decoupled weight decay).
Batch Size	32 images per batch.
Training Process	• Feature Extraction: 4 epochs training only the custom head. • Fine-Tuning: 20 epochs training the entire model.
Training Techniques	• AMP (Automatic Mixed Precision): Enabled to speed up training. • Gradient Clipping: Set to 0.5 to prevent exploding gradients. • Learning Rate Scheduling: ReduceLROnPlateau. • Early Stopping: Used to prevent overfitting.

Important Results

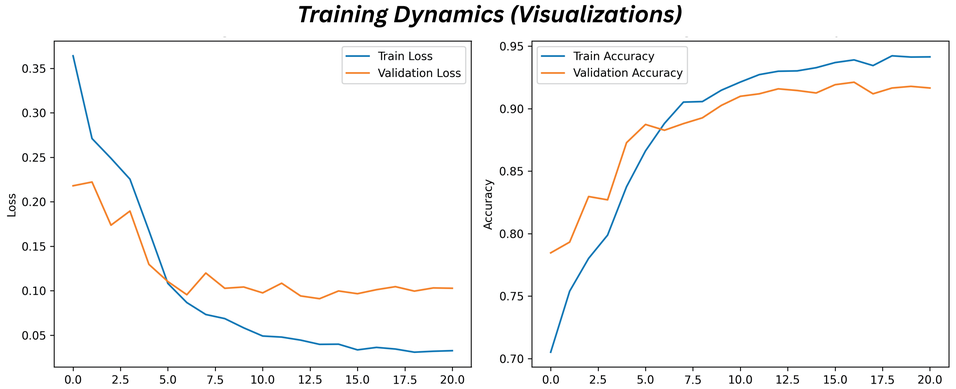
The Vision Transformer (ViT) model achieved a strong overall accuracy of 92.3% on the test set. It demonstrates excellent performance for "Caries," "Mouth Ulcers," and "Tooth Discoloration" with near-perfect F1-scores. The primary area for improvement lies in distinguishing between "Calculus" and "Gingivitis," where the model shows some confusion.

Class	Precision	Recall	F1-Score	Support
Calculus	0.79	0.69	0.74	195
Cariers	0.98	0.98	0.98	296
Gingivitis	0.84	0.91	0.87	353
Healthy Tee	0.85	0.79	0.81	14
Mouth Ulcer	0.99	0.99	0.99	382
Tooth Discoloratic	0.97	0.97	0.97	276
Macro Avg	0.9	0.89	0.89	1516
Weighted Av	0.92	0.92	0.92	1516

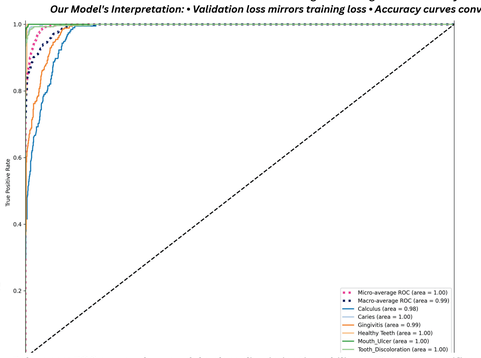
Interpretation:

- **Excellent Performance (F1 > 0.95)**: The model is exceptionally reliable at identifying Caries, Mouth Ulcers, and Tooth Discoloration.
- **Good Performance (F1 0.70 - 0.95)**: The model is effective at detecting Gingivitis (0.87) and Calculus (0.74). The high recall for Gingivitis (0.91) is valuable as it is unlikely to miss positive cases.
- **Areas for Improvement**: The main weakness is the confusion between "Calculus" and "Gingivitis." Additionally, the small sample size for "Healthy Teeth" (14) makes its result less reliable and requires more data.

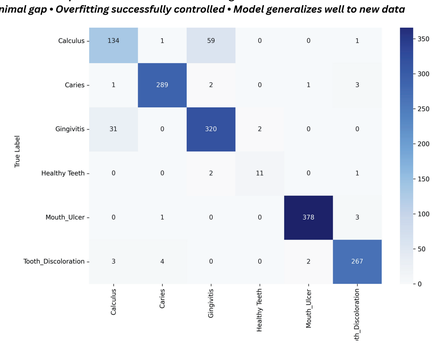
Training Dynamics (Visualizations)



Relevance: Learning curves diagnose model fit by tracking training vs validation performance to detect overfitting.
Our Model's Interpretation: • Validation loss mirrors training loss • Accuracy curves converge with minimal gap • Overfitting successfully controlled • Model generalizes well to new data



Relevance: ROC curves show model's class discrimination ability. AUC score quantifies this (1.0 = perfect classifier).
Our Model's Interpretation: • Curves pushed to top-left corner • All classes have AUC ≈ 1.0 • Excellent discrimination between dental conditions across all thresholds



Relevance: Confusion matrix shows detailed classification breakdown and specific error patterns between classes.
Our Model's Interpretation: • Strong diagonal confirms high overall accuracy • Main weakness: 59 "Calculus" cases misclassified as "Gingivitis" • Error likely due to visual similarities between conditions

Acknowledgement

Grateful to supervising faculty, clinical experts for annotations and validation, and the open-source communities (PyTorch, FastAPI, MongoDB). This work is for educational and research purposes only and is not a substitute for professional diagnosis.

References

- Dosovitskiy et al. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale (ICLR 2021).
- Lin et al. Focal Loss for Dense Object Detection (ICCV 2017).
- He et al. Deep Residual Learning for Image Recognition (CVPR 2016).
- Mendeley Data, Kaggle
- FastAPI, PyTorch, and MongoDB Documentation.

Conclusion

The ViT model offers significant potential for oral disease diagnosis, effectively identifying conditions like Caries, Mouth Ulcers, and Tooth Discoloration. Its main medical challenge lies in distinguishing between Calculus and Gingivitis, which is crucial for accurate treatment. Improving this differentiation is key for practical clinical application.